dimensional[35]), so it can safely stay out of Philippe's way either by slithering ahead of him, as we initially envisioned, or by crawling around the tiny circular dimension and letting Philippe pass above.

The tightrope illustrates that dimensions—the independent directions in which anything can move—come in two qualitatively distinct varieties. They can be big and easy to see, like the left/right dimension of the tightrope's surface, or they can be tiny and more difficult to see, like the clockwise/counterclockwise dimension that circles around the tightrope's surface. In this example, it was not a major challenge to see the small circular girth of the tightrope's surface. All we needed was a reasonable magnifying instrument. But as you can imagine, the smaller a curled-up dimension, the more difficult it is to detect. At a distance of a few miles, it's one thing to reveal the circular dimension of a tightrope's surface; it would be quite another to reveal the circular dimension of something as thin as dental floss or a narrow nerve fiber.

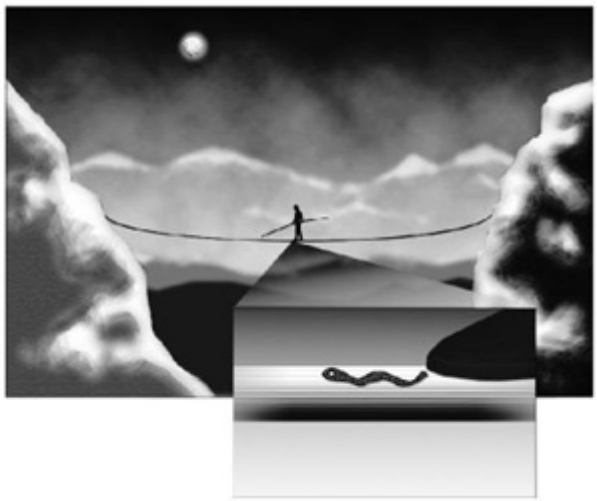


Figure 12.5 From a distance, a tightrope wire looks one-dimensional, although with a strong enough telescope, its second, curled-up dimension becomes visible.

Klein's contribution was to suggest that what's true for an object *within* the universe might be true for the fabric of the universe itself. Namely, just as the tightrope's surface has both large and small dimensions, so does the fabric of space. Maybe the three dimensions we all know about—left/right, back/forth, and up/down—are like the horizontal extent of the tightrope, dimensions of the big, easy-to-see